UTS UNIVERSITY OF TECHNOLOGY SYDNEY

Faculty of Engineering and IT

Assignment Cover Sheet

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Assignment Title: Team Project Report: Frequency-Domain and State-Feedback Control of the QUBE-Servo 2 Rotary Inverted Pendulum

Student Name(s) and Number(s) **Tutorial Group:** 02

Jaylen Avtarovski	24767939	Contribution: [TODO: %]
Rhyse Williams	24817532	Contribution: [TODO: %]
Elina Kim	24949885	Contribution: [TODO: %]

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Statement of Collaboration:
[TODO: The three team members above collaborated on all sections of this report and on the design, simulation and hardware testing of both control systems. Edit to describe how the team actually worked.]

GenAI Declaration:
[TODO: Declare every generative-AI tool used and what it was used for (for example drafting text, LaTeX formatting, MATLAB plotting scripts), following the subject coordinator's advice. This must be accurate.]

Signature [TODO: initials] **Date** [TODO: date]

Individual contribution levels. As permitted by the assessment criteria, the team states each member's contribution level, which scales the individual report mark (individual mark = group mark × contribution level).

Student	Contribution level	Main activities
Jaylen Avtarovski	[TODO: %]	[TODO:]
Rhyse Williams	[TODO: %]	[TODO:]
Elina Kim	[TODO: %]	[TODO:]

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1 Introduction

The rotary inverted pendulum is a benchmark problem in control engineering. A motor-driven horizontal arm must hold a freely hinged pendulum upright, an open-loop unstable equilibrium, while also tracking a commanded arm angle [2, 3]. This project uses the Quanser QUBE-Servo 2 with a Teensy 4.1 microcontroller running the UTS ControLab Simulink template [9, 11].

The team designed, simulated and implemented two control systems that each govern the whole system:

1. **CS1, frequency domain.** Two transfer-function controllers, $C_\theta(s)$ for the arm and $C_\alpha(s)$ for the pendulum, plus a reference prefilter $F(s)$. They are designed in the s-domain and discretised to $C(z)$ for implementation.
2. **CS2, time domain.** A discrete-time state-feedback controller (SFC) with integral action and a Luenberger observer, designed from the discrete state-space model.

Both controllers run inside a single MATLAB Function block at $T_s = 1$ ms. The same block is used by the linear simulation, the nonlinear simulation and the code generated for the Teensy, so the code that was simulated is the code that ran on the QUBE.

Report structure. Section 2 reviews the background, Section 3 states the control objectives, Section 4 develops and verifies the model, and Section 5 covers sensor calibration. Sections 6 and 7 present the design, implementation and results of each control system. Section 8 compares the two designs, Section 9 responds to the feedback from the final demonstration, and Section 10 concludes.

2 Literature review

Furuta *et al.* [2] introduced the rotary pendulum as a compact alternative to the cart–pole. It shares the cart–pole's non-minimum-phase, underactuated dynamics but needs only a rotary joint and no rail. Åström and Furuta [3] later separated the problem into energy-based swing-up and linear stabilisation about the upright equilibrium. This project addresses the second part: the pendulum is raised by hand and the controller catches it near upright.

Classical frequency-domain design shapes the open-loop transfer function so that the closed loop meets bandwidth and robustness targets, which are expressed as gain and phase margins and the peak sensitivity M_s [4, 6]. For an open-loop unstable plant, the Nyquist criterion requires the loop to encircle -1 once for each unstable pole. The gain margin therefore becomes a *band* of admissible gains with both a lower and an upper limit.

State-space design instead places all closed-loop eigenvalues directly by full-state feedback. That needs the velocities, which are not measured on the QUBE, so they are reconstructed by an observer [1, 6]. For digital implementation, the continuous designs are mapped to the z-domain by Tustin or zero-order-hold equivalents, and continuous poles map to $z = e^{sT_s}$ [5, 8].

Inverted-pendulum stabilisation is the core problem in self-balancing transporters, legged and humanoid robot balance, rocket attitude control during vertical launch, and crane anti-sway control. This is why it is so widely used as a teaching and research benchmark [7].

3 Control objectives

Each control system must keep the pendulum upright ($\alpha \to 0$) and move the arm to a commanded angle r. Table 1 lists the targets. The arm step used to assess every design is $r = 45°$. Its $\pm 2\%$ settling band ($\pm 0.9°$) is about five encoder counts wide ($0.176°$ per count). For a $5°$ step the band would be smaller than two counts, so the settling time could not be measured reliably.

Table 1: Closed-loop performance requirements.

Quantity	Requirement	Reason
Arm overshoot M_p	$< 15\%$	Project specification
Arm settling time t_s (2%)	< 5 s	Project specification
Steady-state accuracy	**[TODO:** confirm the specification]	Project specification
Motor voltage $\lvert V_m \rvert$	≤ 10 V	Amplifier limit [9]
Pendulum angle $\lvert \alpha \rvert$	small, about $< 15°$	Keeps the linear model valid

4 System modelling and model verification

The model was derived in Demonstration Practical 2 and is reused here unchanged, so both controllers are designed from the same plant.

4.1 Nonlinear model and linearisation

Definitions and assumptions. The QUBE-Servo 2 rotary pendulum comprises a motor-driven horizontal arm and a pendulum hinged at its tip. Encoders measure the two angles [9]. Let θ denote arm angle and α pendulum angle from the upward vertical, with $\alpha = 0$ upright. Use radians. Let $\boldsymbol{e}_r$ point along the arm, $\boldsymbol{e}_t$ point in the direction of increasing θ and $\boldsymbol{e}_z$ point upward. Positive α tilts the pendulum towards $-\boldsymbol{e}_t$. This explicitly fixes the coupling signs.

The arm and pendulum are approximated as rigid uniform slender rods. The pendulum centre of mass is at $l_p = L_p/2$. Viscous damping represents friction. Motor inductance, structural flexibility, nonlinear friction and pendulum longitudinal-axis inertia are neglected. The motor is direct drive. These are nominal modelling assumptions, not measurements of the particular laboratory unit.

$$x = \begin{bmatrix} \theta & \alpha & \dot{\theta} & \dot{\alpha} \end{bmatrix}^T, \qquad u = V_m, \qquad y = \begin{bmatrix} \theta & \alpha \end{bmatrix}^T. \tag{1}$$

Thus there are $n = 4$ states, $m = 1$ input and $p = 2$ outputs. Velocities are states even though only the angles are directly measured.

Motor electrical dynamics. Kirchhoff's voltage law and the motor relations give

$$V_m = R_m i_m + L_m \frac{di_m}{dt} + e_b, \tag{2}$$

$$e_b = k_m \dot{\theta}, \qquad \tau_m = k_t i_m. \tag{3}$$

Neglecting inductance,

$$i_m = \frac{V_m - k_m \dot{\theta}}{R_m}, \qquad \boxed{\tau_m = \frac{k_t}{R_m} V_m - \frac{k_t k_m}{R_m} \dot{\theta}.} \tag{4}$$

Voltage generates torque and back emf reduces the current as speed rises.

Inertias and energy. For a uniform rod, $dm = (m/L)\,dr$. The relevant inertias are

$$J_{\text{arm}} = \int_0^{L_r} r^2 \frac{m_r}{L_r}\,dr = \frac{1}{3} m_r L_r^2, \qquad J_r = J_{\text{arm}} + J_m + J_h, \tag{5}$$

$$J_p = \int_{-L_p/2}^{L_p/2} r^2 \frac{m_p}{L_p}\,dr = \frac{1}{12} m_p L_p^2. \tag{6}$$

Here J_r includes the arm, rotor and hub. The arm rod approximation excludes the separately added rotor and hub. No inertia-disc module is included. J_p is about the pendulum centre of mass; the hinge inertia is $J_p + m_p l_p^2 = \frac{1}{3} m_p L_p^2$ by the parallel-axis theorem.

The pendulum centre-of-mass position is

$$\boldsymbol{r}_p = L_r \boldsymbol{e}_r - l_p \sin\alpha\, \boldsymbol{e}_t + l_p \cos\alpha\, \boldsymbol{e}_z. \tag{7}$$

Using $\dot{\boldsymbol{e}}_r = \dot\theta \boldsymbol{e}_t$ and $\dot{\boldsymbol{e}}_t = -\dot\theta \boldsymbol{e}_r$ gives

$$\dot{\boldsymbol{r}}_p = l_p \sin\alpha\, \dot\theta\, \boldsymbol{e}_r + (L_r\dot\theta - l_p \cos\alpha\, \dot\alpha)\boldsymbol{e}_t - l_p \sin\alpha\, \dot\alpha\, \boldsymbol{e}_z, \tag{8}$$

$$\|\dot{\boldsymbol{r}}_p\|^2 = (L_r^2 + l_p^2 \sin^2\alpha)\dot\theta^2 + l_p^2 \dot\alpha^2 - 2 L_r l_p \cos\alpha\, \dot\theta\dot\alpha. \tag{9}$$

Neglecting longitudinal-axis inertia, pendulum rotation about its centre of mass contributes

$$\frac{1}{2} J_p(\dot\alpha^2 + \sin^2\alpha\, \dot\theta^2).$$

Therefore the total kinetic and potential energies are

$$T = \frac{1}{2} J_r \dot\theta^2 + \frac{1}{2} m_p \|\dot{\boldsymbol{r}}_p\|^2 + \frac{1}{2} J_p(\dot\alpha^2 + \sin^2\alpha\, \dot\theta^2), \tag{10}$$

$$U = m_p g l_p \cos\alpha. \tag{11}$$

Potential energy is referenced to the hinge height and is maximal upright.

Define

$$a = J_r + m_p L_r^2, \quad b = J_p + m_p l_p^2, \quad c = m_p L_r l_p, \quad h = m_p g l_p. \tag{12}$$

The Lagrangian is

$$\mathcal{L} = T - U = \frac{1}{2}(a + b\sin^2\alpha)\dot\theta^2 + \frac{1}{2} b\dot\alpha^2 - c\cos\alpha\, \dot\theta\dot\alpha - h\cos\alpha. \tag{13}$$

Euler–Lagrange equations. For $q \in \{\theta, \alpha\}$,

$$\frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot q}\right) - \frac{\partial \mathcal{L}}{\partial q} = Q_q, \qquad Q_\theta = \tau_m - D_r\dot\theta, \quad Q_\alpha = -D_p\dot\alpha. \tag{14}$$

The arm derivatives are

$$\frac{\partial \mathcal{L}}{\partial \theta} = 0, \tag{15}$$

$$\frac{\partial \mathcal{L}}{\partial \dot\theta} = (a + b\sin^2\alpha)\dot\theta - c\cos\alpha\, \dot\alpha, \tag{16}$$

$$\frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot\theta}\right) = (a + b\sin^2\alpha)\ddot\theta + 2b\sin\alpha\cos\alpha\, \dot\alpha\dot\theta - c\cos\alpha\, \ddot\alpha + c\sin\alpha\, \dot\alpha^2. \tag{17}$$

The pendulum derivatives are

$$\frac{\partial \mathcal{L}}{\partial \dot\alpha} = b\dot\alpha - c\cos\alpha\, \dot\theta, \tag{18}$$

$$\frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot{\alpha}}\right) = b\ddot{\alpha} - c\cos\alpha\,\ddot{\theta} + c\sin\alpha\,\dot{\alpha}\dot{\theta}, \tag{19}$$

$$\frac{\partial \mathcal{L}}{\partial \alpha} = b\sin\alpha\cos\alpha\,\dot{\theta}^2 + c\sin\alpha\,\dot{\theta}\dot{\alpha} + h\sin\alpha. \tag{20}$$

Define $k_v = k_t/R_m$ and $d = D_r + k_t k_m/R_m$. The resulting nonlinear mechanical equations, including motor torque, are

$$(a + b\sin^2\alpha)\ddot{\theta} - c\cos\alpha\,\ddot{\alpha} + 2b\sin\alpha\cos\alpha\,\dot{\alpha}\dot{\theta} + c\sin\alpha\,\dot{\alpha}^2 = k_v u - d\dot{\theta}, \tag{21a}$$

$$-c\cos\alpha\,\ddot{\theta} + b\ddot{\alpha} - b\sin\alpha\cos\alpha\,\dot{\theta}^2 - h\sin\alpha = -D_p\dot{\alpha}. \tag{21b}$$

Fixing the arm and neglecting friction gives $b\ddot{\alpha} = h\sin\alpha$. A small positive tilt therefore accelerates away from upright, which provides a physical check on the gravity sign.

Nonlinear state-space representation. Define the torque expressions

$$r_1(x,u) = k_v u - dx_3 - 2b\sin x_2\cos x_2\,x_4 x_3 - c\sin x_2\,x_4^2, \tag{22}$$

$$r_2(x,u) = b\sin x_2\cos x_2\,x_3^2 + h\sin x_2 - D_p x_4. \tag{23}$$

The equations of motion become

$$\underbrace{\begin{bmatrix} a + b\sin^2\alpha & -c\cos\alpha \\ -c\cos\alpha & b \end{bmatrix}}_{M(\alpha)} \begin{bmatrix} \ddot{\theta} \\ \ddot{\alpha} \end{bmatrix} = \begin{bmatrix} r_1 \\ r_2 \end{bmatrix}. \tag{24}$$

The mass-matrix determinant and inverse are

$$\Delta(\alpha) = b(a + b\sin^2\alpha) - c^2\cos^2\alpha, \tag{25}$$

$$M^{-1}(\alpha) = \frac{1}{\Delta(\alpha)}\begin{bmatrix} b & c\cos\alpha \\ c\cos\alpha & a + b\sin^2\alpha \end{bmatrix}. \tag{26}$$

Hence

$$\boxed{\dot{x} = f(x,u) = \begin{bmatrix} x_3 \\ x_4 \\ (br_1 + c\cos x_2\,r_2)/\Delta(x_2) \\ (c\cos x_2\,r_1 + (a + b\sin^2 x_2)r_2)/\Delta(x_2) \end{bmatrix}}. \tag{27}$$

For the parameters in Section 4, $ab - c^2 > 0$ and $\Delta(\alpha) = (ab - c^2) + (b^2 + c^2)\sin^2\alpha > 0$, so this idealised model is nonsingular for every pendulum angle.

Equilibrium and Jacobian linearisation. An upright stationary equilibrium is

$$x_e = \begin{bmatrix} \theta_e & 0 & 0 & 0 \end{bmatrix}^T, \qquad u_e = 0. \tag{28}$$

Any fixed θ_e is an ideal equilibrium, since no restoring torque depends on arm angle. Choose $\theta_e = 0$ for the numerical model. Define $\delta x = x - x_e$, $\delta u = u - u_e$ and $\delta y = y - y_e$. The first-order Taylor expansion is

$$\delta\dot{x} = A_c\delta x + B_c\delta u, \qquad \delta y = C_c\delta x + D_c\delta u, \tag{29}$$

with Jacobians

$$A_c = \left.\frac{\partial f}{\partial x}\right|_e, \qquad B_c = \left.\frac{\partial f}{\partial u}\right|_e. \tag{30}$$

At equilibrium, $r_1 = r_2 = 0$ and $\Delta_0 = ab - c^2$. Also,

$$\left.\frac{\partial(r_1, r_2)}{\partial x}\right|_e = \begin{bmatrix} 0 & 0 & -d & 0 \\ 0 & h & 0 & -D_p \end{bmatrix}, \qquad \left.\frac{\partial(r_1, r_2)}{\partial u}\right|_e = \begin{bmatrix} k_v \\ 0 \end{bmatrix}. \tag{31}$$

Derivatives of M^{-1} multiply the zero equilibrium torque vector. Multiplying the above derivatives by $M^{-1}(0) = \Delta_0^{-1}\begin{bmatrix} b & c \\ c & a \end{bmatrix}$ gives

$$\boxed{A_c = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & ch/\Delta_0 & -bd/\Delta_0 & -cD_p/\Delta_0 \\ 0 & ah/\Delta_0 & -cd/\Delta_0 & -aD_p/\Delta_0 \end{bmatrix}, \quad B_c = \begin{bmatrix} 0 \\ 0 \\ bk_v/\Delta_0 \\ ck_v/\Delta_0 \end{bmatrix}.} \tag{32}$$

Since the outputs are the two angles with no direct voltage feedthrough,

$$\boxed{C_c = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}, \qquad D_c = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.} \tag{33}$$

This linearisation describes small deviations about upright; large-angle motion requires the nonlinear model in (27).

4.2 System parameters

Table 2 lists the manufacturer-supplied parameters relevant to the pendulum configuration. The inertia-disc parameters are excluded. Motor inductance is listed to justify its neglect, rather than being retained as a model state.

Table 2: Parameters from Table 2.2 of the QUBE-Servo 2 user manual [9].

Symbol	Description	Value
R_m	Motor resistance	$8.4\,\Omega$
k_t	Torque constant	$0.042\,\mathrm{N\,m/A}$
k_m	Back-emf constant	$0.042\,\mathrm{V\,s/rad}$
J_m	Motor rotor inertia	$4.0 \times 10^{-6}\,\mathrm{kg\,m^2}$
J_h	Attachment hub inertia	$0.6 \times 10^{-6}\,\mathrm{kg\,m^2}$
L_m	Inductance (neglected)	$1.16 \times 10^{-3}\,\mathrm{H}$
m_r	Arm mass	$0.095\,\mathrm{kg}$
L_r	Arm length, motor pivot to tip	$0.085\,\mathrm{m}$
m_p	Pendulum mass	$0.024\,\mathrm{kg}$
L_p	Pendulum full length	$0.129\,\mathrm{m}$
	Encoder resolution (quadrature)	$2\pi/2048\,\mathrm{rad/count}$
	Recommended amplifier output	$\pm 10\,\mathrm{V}$

The motor electrical time constant is

$$\tau_{\mathrm{elec}} = \frac{L_m}{R_m} = \frac{1.16 \times 10^{-3}}{8.4} = 1.3810 \times 10^{-4}\,\mathrm{s}. \tag{34}$$

This is short compared with the mechanical time scales, motivating the quasi-static current approximation. Retaining inductance would introduce motor current as a fifth state.

The damping values in Table 3 are adopted from the manufacturer's student workbook, *Pendulum State-Space Model*, in-lab exercise 6 [10]. That source reports them as experimentally obtained example values; they were not identified from this group's specific laboratory unit. No new experimental identification is claimed.

Table 3: Additional constants and their provenance.

Symbol	Description	Value	Basis
D_r	Arm damping	$0.0015\,\mathrm{N\,m\,s/rad}$	Workbook example
D_p	Pendulum damping	$0.0005\,\mathrm{N\,m\,s/rad}$	Workbook example
g	Gravity	$9.81\,\mathrm{m/s^2}$	Standard approximation

The derived dimensions and inertias are

$$l_p = L_p/2 = 0.0645\,\mathrm{m}, \tag{35}$$
$$J_{\mathrm{arm}} = \tfrac{1}{3}(0.095)(0.085)^2 = 2.2879167 \times 10^{-4}\,\mathrm{kg\,m^2}, \tag{36}$$
$$J_r = J_{\mathrm{arm}} + 4.0 \times 10^{-6} + 0.6 \times 10^{-6} = 2.3339167 \times 10^{-4}\,\mathrm{kg\,m^2}, \tag{37}$$
$$J_p = \tfrac{1}{12}(0.024)(0.129)^2 = 3.3282 \times 10^{-5}\,\mathrm{kg\,m^2}. \tag{38}$$

The constants used in the matrices are

$$a = 4.0679167 \times 10^{-4}\,\mathrm{kg\,m^2}, \qquad b = 1.33128 \times 10^{-4}\,\mathrm{kg\,m^2}, \tag{39}$$
$$c = 1.3158 \times 10^{-4}\,\mathrm{kg\,m^2}, \qquad h = 0.01518588\,\mathrm{N\,m}, \tag{40}$$
$$d = 0.00171\,\mathrm{N\,m\,s/rad}, \qquad k_v = 0.005\,\mathrm{N\,m/V}, \tag{41}$$
$$\Delta_0 = 3.68420646 \times 10^{-8}\,\mathrm{kg^2\,m^4}. \tag{42}$$

Substituting into (32) gives

$$A_c \simeq \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 54.2357795 & -6.17904785 & -1.78573054 \\ 0 & 167.674898 & -6.10719845 & -5.52075014 \end{bmatrix}, \quad B_c \simeq \begin{bmatrix} 0 \\ 0 \\ 18.0673914 \\ 17.8573054 \end{bmatrix}. \tag{43}$$

The matrices C_c, D_c remain exactly those in (33). These are nominal simulation parameters. Actual mass distribution, damping, encoder calibration and unmodelled friction may produce differences from hardware measurements.

4.3 Discretisation and sampling time

Choose the sampling time

$$\boxed{T_s = 0.001\,\mathrm{s}, \qquad f_s = 1/T_s = 1000\,\mathrm{Hz}.} \tag{44}$$

This matches the value in the supplied ControLab template's `Hardware_settings.m` [11]. It is also short relative to the mechanical dynamics, since $T_s \max_i |\lambda_i(A_c)| \simeq 0.01763 \ll 1$ for this model. Hardware execution timing and achieved closed-loop performance must still be checked during implementation.

Zero-order hold (ZOH) assumes that motor voltage is held constant between controller updates. The exact discretisation of the linearised model under this assumption is

$$A = e^{A_c T_s}, \qquad B = \int_0^{T_s} e^{A_c \tau} B_c \, d\tau, \qquad C = C_c, \quad D = D_c. \tag{45}$$

The MATLAB function `c2d` evaluates these expressions without requiring an inverse of A_c. This matters because the arm-position mode makes A_c singular. The discrete-time model is

$$\delta x[k+1] = A\delta x[k] + B\delta u[k], \tag{46a}$$
$$\delta y[k] = C\delta x[k] + D\delta u[k]. \tag{46b}$$

At the chosen zero equilibrium the deviation variables equal the original variables. The calculated matrices are

$$A \simeq \begin{bmatrix} 1 & 2.70127657\times10^{-5} & 9.96918625\times10^{-4} & -8.80391044\times10^{-7} \\ 0 & 1.00008362958 & -3.04176177\times10^{-6} & 9.97274363\times10^{-4} \\ 0 & 0.05392103923 & 0.99384536883 & -0.00174835485 \\ 0 & 0.16705290505 & -0.00607175726 & 0.99458335877 \end{bmatrix}, \tag{47}$$

$$B \simeq \begin{bmatrix} 9.00986699\times10^{-6} \\ 8.89404027\times10^{-6} \\ 0.0179959975588 \\ 0.0177536761881 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}, \quad D = \begin{bmatrix} 0 \\ 0 \end{bmatrix}. \tag{48}$$

The dimensions are $A \in \mathbb{R}^{4\times4}$, $B \in \mathbb{R}^{4\times1}$, $C \in \mathbb{R}^{2\times4}$ and $D \in \mathbb{R}^{2\times1}$. Displayed results are rounded; full precision must be retained in MATLAB for controller and observer design.

4.4 Transfer functions for frequency-domain design

From $G(s) = C_c(sI - A_c)^{-1}B_c$ (MATLAB `tf(ss(Ac,Bc,Cc,Dc))`), both outputs share the denominator

$$D(s) = s\,(s^3 + d_3s^2 + d_2s + d_1) = s^4 + 11.70s^3 - 144.47s^2 - 704.84s, \tag{49}$$

where $d_3 = (aD_p + bd)/\Delta_0$, $d_2 = (dD_p - ah)/\Delta_0$ and $d_1 = -dh/\Delta_0$. The two transfer functions are

$$G_\theta(s) = \frac{\theta(s)}{V_m(s)} = \frac{18.07s^2 + 67.86s - 2060.9}{D(s)} = \frac{18.07\,(s-8.97)(s+12.72)}{s\,(s-9.95)(s+4.02)(s+17.6)}, \tag{50}$$

$$G_\alpha(s) = \frac{\alpha(s)}{V_m(s)} = \frac{17.86\,s^2}{D(s)} = \frac{17.86\,s}{(s-9.95)(s+4.02)(s+17.6)}. \tag{51}$$

These transfer functions show three properties that drive the design:

- **Open-loop unstable.** The pole at $s = +9.95\ \mathrm{s}^{-1}$ is the falling pendulum (time constant about 0.1 s), and the pole at $s = 0$ reflects that the arm has no restoring force. A stabilising pendulum loop is mandatory.
- **Non-minimum phase arm.** G_θ has a right-half-plane zero at $s = +8.97$. To move the arm positively, the base must first swing the wrong way to tilt the pendulum. This limits the achievable arm bandwidth and causes the initial undershoot seen in the responses.
- **Zero at $s = 0$ in G_α.** A constant voltage cannot hold a steady tilt, so the pendulum loop alone cannot fix the arm position. An outer arm loop is required.

4.5 Model verification

The linear model was verified against the nonlinear model by simulating both with the same controller code. For the 45° arm step, the pendulum stays within 2.4° (CS1) and 4.6° (CS2) of upright. The linear and nonlinear responses overlap within plotting accuracy (Figures 6 and 11), which confirms that the small-angle assumption holds over the operating range.

Model signs were checked on the hardware. With the controller disabled and a small positive voltage applied, θ increased as the model predicts. The pendulum encoder measured α in the opposite direction to the model's convention, so a gain of -1 was placed before the controller (Section 5). After this correction the CS1 controller balanced the pendulum on its first attempt, which independently confirms the sign structure of (50)–(51). **[TODO:** If the team ran an open-loop or system-identification test, add a measured-versus-model plot here. The DP2 damping values are textbook estimates, not identified from this QUBE.]

5 Calibration of transducers and sensors

Both angles are measured by quadrature encoders with 2048 counts per revolution, a resolution of $2\pi/2048 = 3.07$ mrad ($0.176°$) per count [9]. The calibration steps were:

1. **Zero of α.** The pendulum encoder counts from the angle at power-up. The pendulum is therefore started hanging down, where the template measures $\alpha = -\pi$, and the template's offset makes upright $\alpha = 0$, as the model requires. Each run starts by checking that the hanging reading is $-180°$.
2. **Direction of α.** A gain of -1 makes a positive tilt match the model's $-\boldsymbol{e}_t$ convention (Section 4.5).
3. **Direction and zero of θ.** The arm angle is zero at power-up. A positive voltage increases the measured θ, matching the model, so no correction was needed.
4. **Motor voltage.** The commanded V_m is saturated to ± 10 V inside the controller block, so the observer and the logged data always see the voltage the amplifier actually applies.
5. **Encoder read errors.** The logged arm angle contains occasional single-sample spikes of the order of 10^5 degrees (2 in run 04). These are SPI read errors, not motion, and they are removed before plotting by holding the previous sample whenever θ jumps by more than $90°$ in 1 ms.

[TODO: Describe any further calibration the team did, for example checking the encoder scale against a protractor at $\pm 90°$.**]**

6 Control System 1: frequency-domain design

6.1 Control structure and why it was chosen

Figure 1 shows the structure. The plant has one input and two outputs, so a single SISO loop cannot both stabilise α and fix θ (Section 4). Two controllers act on one voltage:

$$V_m = \mathrm{sat}_{\pm 10}\left[C_\theta(s)\big(F(s)r - \theta\big) - C_\alpha(s)\,\alpha\right]. \tag{52}$$

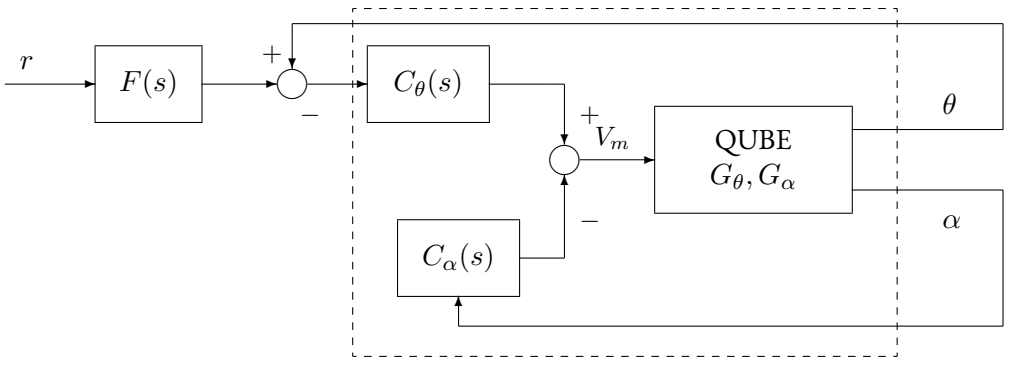


Figure 1: CS1 structure. Everything inside the dashed box runs in one MATLAB Function block at $T_s = 1$ ms. The saturation is omitted for clarity.

Controller type. Both C_θ and C_α are filtered PD controllers:

$$C_i(s) = K_{p,i} + K_{d,i}\frac{Ns}{s+N}, \qquad i \in \{\theta, \alpha\}, \qquad N = 60\,\mathrm{rad/s}. \tag{53}$$

- *Derivative action* adds the phase lead needed to stabilise the unstable pole at $s = +9.95$. A proportional-only pendulum loop cannot do this, because the gravity term makes the loop phase too negative near crossover.

- *The first-order filter* at $N = 60$ rad/s limits the gain on encoder quantisation noise. An ideal derivative of a signal that changes in 0.176° steps every few milliseconds would drive the voltage between its limits. N is placed about $3\times$ above the target crossover of about 20 rad/s, so the filter costs only about 18° of phase there.
- *No integrator* is needed. The plant already contains the free integrator $1/s$ from arm angle to voltage, so the arm loop is type 1 and has zero steady-state error to a step without friction.

6.2 Design method

Step 1: target closed-loop poles from the specifications. The dominant pair is chosen from the second-order relations $M_p = e^{-\pi\zeta/\sqrt{1-\zeta^2}}$ and $t_s \approx 4/(\zeta\omega_n)$:

$$\zeta = 0.7 \Rightarrow M_p = 4.6\% < 15\%, \qquad \omega_n = 8\,\text{rad/s} \Rightarrow t_s \approx \frac{4}{0.7 \times 8} = 0.71\,\text{s} < 5\,\text{s}. \tag{54}$$

This leaves margin for the RHP zero, friction and quantisation, which all slow and degrade the real response. The bandwidth was *not* set by the settling-time limit alone. An earlier, slower design ($\omega_n = 2.5$, $\zeta = 0.6$) met the specifications in simulation but showed a persistent arm oscillation once encoder quantisation was modelled. Its pendulum loop had too little gain to reject the quantised measurement. The faster pole set raises the pendulum-loop gain while keeping the voltage small.

Two further poles are placed at $-2.4\omega_n = -19.2$ and $-3.2\omega_n = -25.6$, at least $3\times$ faster than the dominant pair, so they add little to the transient.

Step 2: characteristic polynomial. With (53) written as $C_i(s) = \big[(K_{p,i}+K_{d,i}N)s+K_{p,i}N\big]/(s+N)$ and the plant numerators $n_\theta(s) = n_2s^2 + n_1s + n_0$ and $n_\alpha(s) = m_2s^2$, the closed-loop characteristic equation of (52) is

$$D(s)(s+N) + n_\theta(s)\big[(K_{p\theta} + K_{d\theta}N)s + K_{p\theta}N\big] + n_\alpha(s)\big[(K_{p\alpha} + K_{d\alpha}N)s + K_{p\alpha}N\big] = 0. \tag{55}$$

This polynomial has degree five, and its s^5 and s^4 coefficients (1 and $N + d_3 = 71.70$) do not depend on the gains. The sum of the five closed-loop poles is therefore fixed at $-(N + d_3)$, so the fifth pole is *not* free:

$$p_5 = -(N + d_3) - \big(2(-5.6) - 19.2 - 25.6\big) = -15.7\,\text{s}^{-1}. \tag{56}$$

This pole is still faster than the dominant pair.

Step 3: solve for the gains. Matching the s^3, s^2, s^1, s^0 coefficients of (55) with $\prod_{k=1}^{5}(s-p_k) = s^5+71.70s^4+1936.5s^3 + 24971s^2 + 162900s + 493873$ gives four linear equations in the four gains:

$$\underbrace{\begin{bmatrix} n_2 & n_2N & m_2 & m_2N \\ n_1 + n_2N & n_1N & m_2N & 0 \\ n_0 + n_1N & n_0N & 0 & 0 \\ n_0N & 0 & 0 & 0 \end{bmatrix}}_{M} \begin{bmatrix} K_{p\theta} \\ K_{d\theta} \\ K_{p\alpha} \\ K_{d\alpha} \end{bmatrix} = \begin{bmatrix} 1936.5 - 557.5 \\ 24971 + 9373 \\ 162900 + 42291 \\ 493873 - 0 \end{bmatrix}, \tag{57}$$

where the right-hand side is the desired coefficient minus the corresponding coefficient of $D(s)(s + N)$. M is nonsingular because $n_0 \neq 0$ and $m_2 \neq 0$: the plant has no pole–zero cancellation that would leave a closed-loop pole uncontrollable by the two loops. Solving with `M\rhs` in MATLAB (`params_CS1A_candidate.m`) gives

$$K_{p\theta} = -3.994, \quad K_{d\theta} = -1.724, \quad K_{p\alpha} = 42.90, \quad K_{d\alpha} = 2.384. \tag{58}$$

- **Arm gains are negative.** The RHP zero makes $n_0 < 0$, so the steady-state arm gain is negative. To move the arm towards $+r$, the arm loop must command a negative voltage first, tilting the pendulum forward.
- **Pendulum loop dominates.** $K_{p\alpha} \gg |K_{p\theta}|$, as expected: the pendulum loop provides stability, and the arm loop is a slow outer correction.

Step 4: reference prefilter. The arm controller adds a closed-loop zero at $z_\theta = -K_{p\theta}N/(K_{p\theta} + K_{d\theta}N) = -2.23\,\mathrm{s}^{-1}$. This is slower than the dominant poles and would cause large overshoot. The prefilter cancels this zero from the reference path and adds a smooth 3 rad/s lag:

$$F(s) = \frac{2.23}{s + 2.23} \cdot \frac{3}{s + 3}, \qquad F(0) = 1. \tag{59}$$

$F(s)$ is outside the feedback loop, so it changes the overshoot and the voltage demand *without moving the closed-loop poles*. This is the standard way to reduce overshoot while keeping the designed stability and disturbance rejection [4].

6.3 Frequency-domain verification

The design is verified on the loop transfer function broken at the motor input, $L(s) = C_\theta(s)G_\theta(s)+C_\alpha(s)G_\alpha(s)$, so that $V_m = -L(s)V_m$ around the loop. This is the single point that both loops share, and where a motor-gain error or an actuator delay enters. $L(s)$ was built in state space (`ss([C_theta_s, C_alpha_s])*G`) because the product of high-order transfer-function polynomials is numerically ill-conditioned.

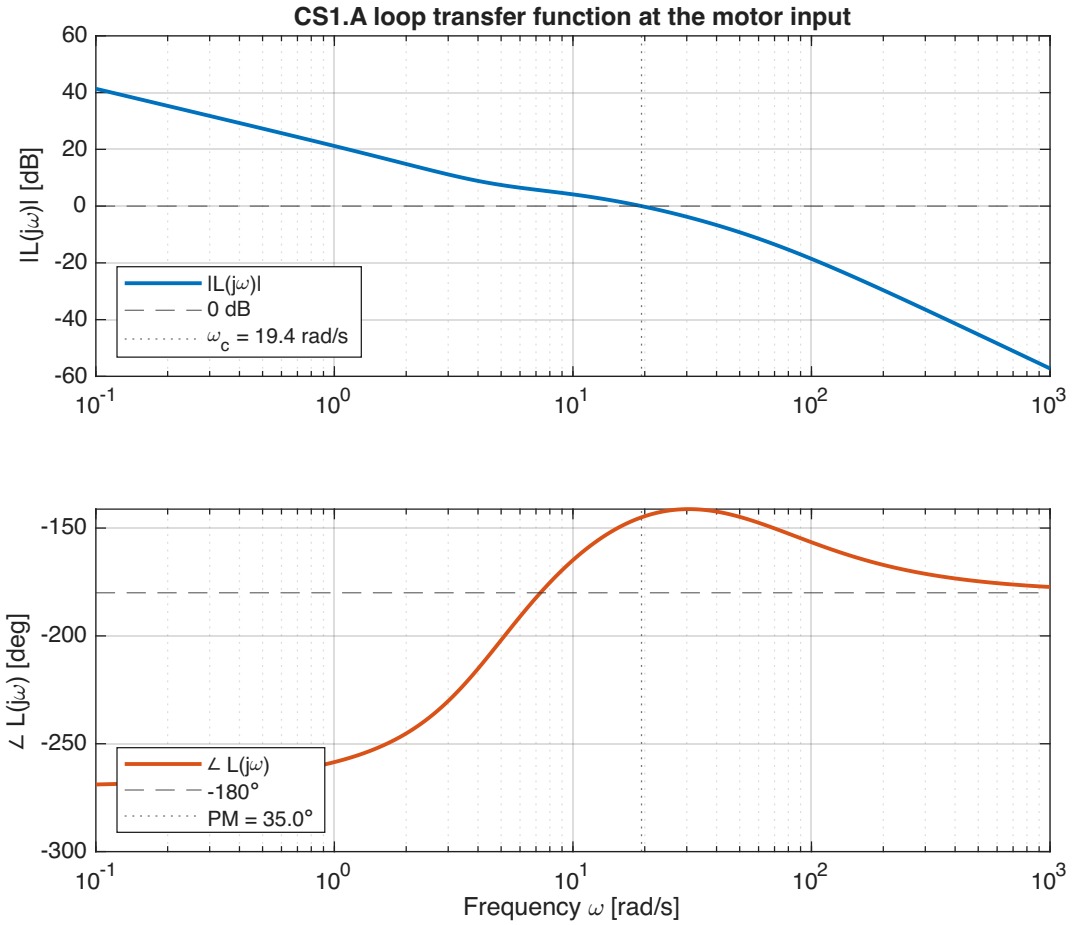


Figure 2: Bode diagram of the CS1 loop $L(j\omega)$ at the motor input. The gain crosses 0 dB at $\omega_c = 19.4$ rad/s with a phase margin of $35.0°$.

Figure 2 shows the loop crossing 0 dB at $\omega_c = 19.4$ rad/s with a phase margin of $35.0°$. The phase starts near $-270°$ and rises through $-180°$ at about 7 rad/s, which is characteristic of a loop that must stabilise an unstable pole:

- By the Nyquist criterion, L has one RHP pole ($P = 1$), so stability requires one anticlockwise encirclement of -1. The phase must cross $-180°$ from below while $|L| > 1$.

- The resulting gain margin is a band:
  - the loop gain may fall to $0.53\times$ nominal before the pendulum is lost;
  - it may rise without limit in the continuous model.
- In the implemented discrete loop (Tustin controller, ZOH plant, $T_s = 1$ ms), the band is 0.53–$68\times$, with a phase margin of $34.4°$ and a delay margin of 31 samples (31 ms).

The lower limit is the practical constraint. Friction at the small balancing voltages reduces the effective motor gain, and the design tolerates a reduction of almost half.

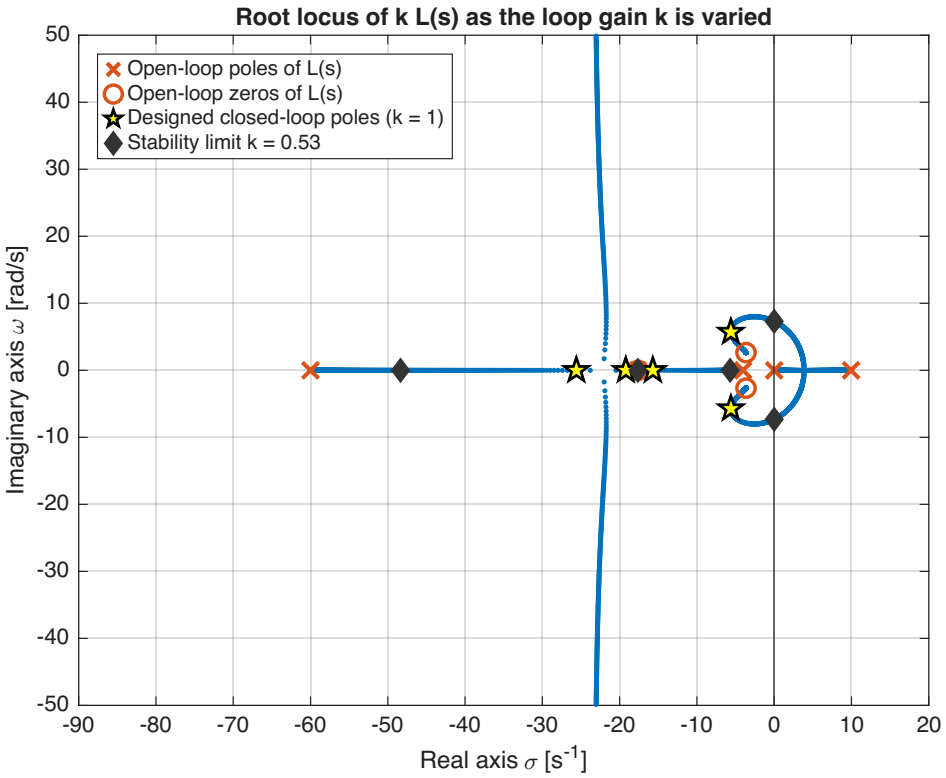


Figure 3: Root locus of $kL(s)$ as the loop-gain multiplier k varies. Stars mark the designed closed-loop poles at $k = 1$. The diamonds mark the poles at $k = 0.53$, where a complex pair is on the imaginary axis.

Figure 3 gives the same information as a root locus:

- As k increases from zero, the branches from the unstable pole $s = +9.95$ and the integrator pole $s = 0$ meet on the real axis and break away as a complex pair. This pair crosses into the left half-plane at $s \approx \pm 7.3j$ when $k = 0.53$, which is the lower gain margin and the $-180°$ crossing frequency of Figure 2.
- At $k = 1$ the five poles sit exactly at the targets of Step 1, confirming the algebra.
- Near $k = 1$, the dominant pair moves only slowly along its branch towards the nearby open-loop zeros, so moderate gain errors change the damping only slightly.

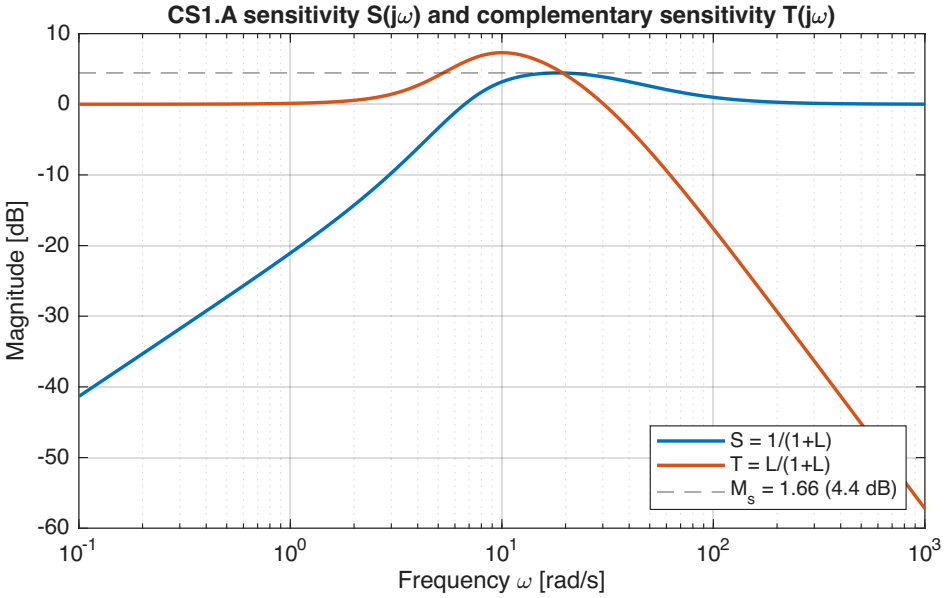


Figure 4: Sensitivity $S(j\omega)$ and complementary sensitivity $T(j\omega)$ of the CS1 loop.

The peak sensitivity is $M_s = 1.66$ (Figure 4), within the usual guideline of $M_s < 2$ [4]. Disturbances below about 5 rad/s, such as a bump or friction, are attenuated. The peak near ω_c amplifies disturbances by at most $1.66\times$. Above 100 rad/s, $|T|$ rolls off, so encoder quantisation noise is attenuated before it reaches the motor.

6.4 Discretisation

The controllers are mapped to the z-domain with the bilinear (Tustin) transform $s \leftarrow \frac{2}{T_s}\frac{z-1}{z+1}$, using MATLAB `c2d(C_s, ts, 'tustin')` [8]:

$$C_\alpha(z) = \frac{181.770 - 179.271z^{-1}}{1 - 0.94175z^{-1}}, \qquad C_\theta(z) = \frac{-104.438 + 104.205z^{-1}}{1 - 0.94175z^{-1}}, \tag{60}$$

$$F(z) = \frac{1.668 \times 10^{-6}(1 + 2z^{-1} + z^{-2})}{1 - 1.99478z^{-1} + 0.99478z^{-2}}. \tag{61}$$

The derivative-filter pole maps to $z = 0.94175$, close to $e^{-NT_s} = 0.94176$. Tustin was chosen for two reasons:

- it maps the stable left half-plane onto the unit disc, so a stable $C(s)$ always gives a stable $C(z)$;
- it preserves the frequency response up to the frequency-warping term, which is negligible here because $\omega_c T_s \approx 0.02$.

Figure 5 confirms that $C_\alpha(z)$ and $C_\alpha(s)$ are indistinguishable up to 1000 rad/s. The sampling rate is 50 times the crossover frequency, and the extra phase lag from the ZOH, about $\omega_c T_s/2 = 0.6°$, is negligible.

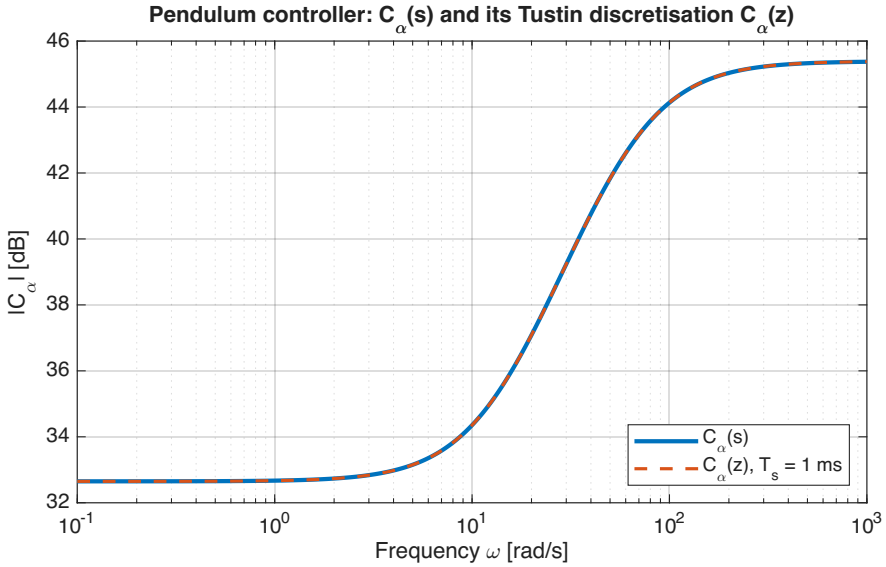


Figure 5: Magnitude of the pendulum controller $C_\alpha(s)$ and of its Tustin discretisation $C_\alpha(z)$ at $T_s = 1$ ms.

6.5 Implementation

The control law (52) runs in one MATLAB Function block (`ctrl_CS1A.slx`). That block is referenced by the linear simulation, the nonlinear simulation and the hardware model, so all three use the same code. Each stage $b(z^{-1})/a(z^{-1})$ is evaluated as a difference equation in direct form II transposed, which needs only n states for an nth-order stage. The block also does the following:

- When the controller is disabled (`reset`=1), it clears every stage state and outputs 0 V, so enabling it causes no voltage jump.
- It saturates V_m to ± 10 V.

The coefficients come from `params_CS1A_candidate.m`, which `Hardware_settings.m` runs before every build.

```
[r_f, wF] = lti_step(bF, aF, r, wF); % F(z)
[u_theta, wT] = lti_step(bT, aT, r_f - theta, wT); % C_theta(z)
[u_alpha, wA] = lti_step(bA, aA, 0 - alpha, wA); % C_alpha(z)
u = min(max(u_theta + u_alpha, -u_max), u_max); % +/-10 V limit

function [y, w] = lti_step(b, a, x, w) % y = b(z^-1)/a(z^-1) x, DF-II T
n = numel(a) - 1; y = b(1)*x;
if n > 0
    y = y + w(1);
    for i = 1:n-1, w(i) = b(i+1)*x + w(i+1) - a(i+1)*y; end
    w(n) = b(n+1)*x - a(n+1)*y;
end
end
```

6.6 Simulation results

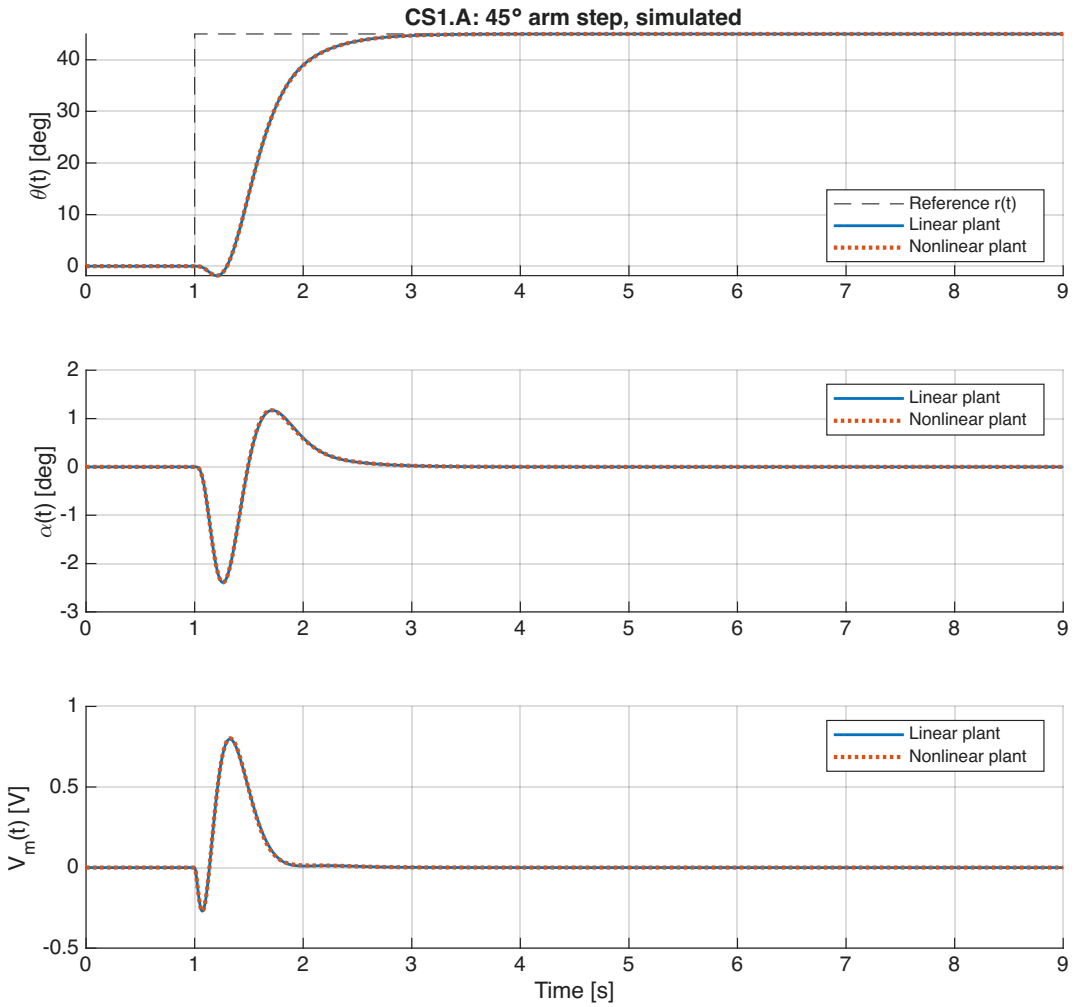


Figure 6: CS1 simulated response to a 45° arm step at $t = 1$ s on the linear and nonlinear plant models: arm angle θ, pendulum angle α and motor voltage V_m.

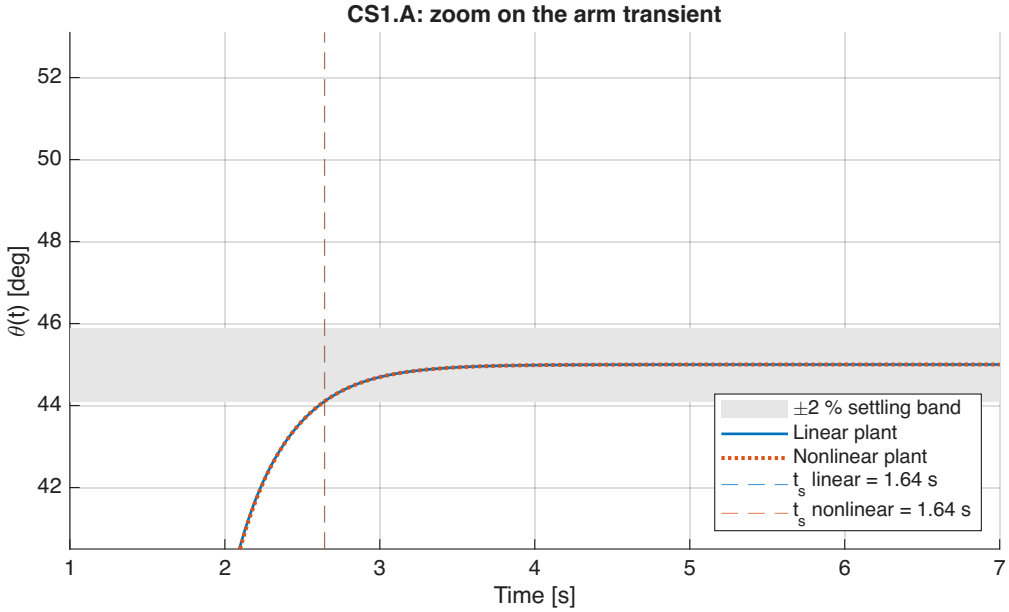


Figure 7: Zoom on the CS1 arm transient of Figure 6, with the $\pm 2\%$ settling band.

Figures 6 and 7 show the response:

- **Initial undershoot.** The arm first moves slightly negative, the signature of the RHP zero. The voltage goes negative first to tilt the pendulum forward.
- **Specifications met.** The arm then rises without overshoot ($M_p = 0.0\%$) and settles within 2% in $t_s = 1.64$ s on the nonlinear plant, well inside both requirements.
- **Prefilter effect.** The settling time is longer than the 0.71 s dominant-pole estimate because the 3 rad/s prefilter lag deliberately slows the *reference*. This keeps the peak voltage at only 0.8 V and the pendulum tilt at $2.4°$.
- **Linear and nonlinear agree.** The two curves coincide, because α stays small.

6.7 Experimental results

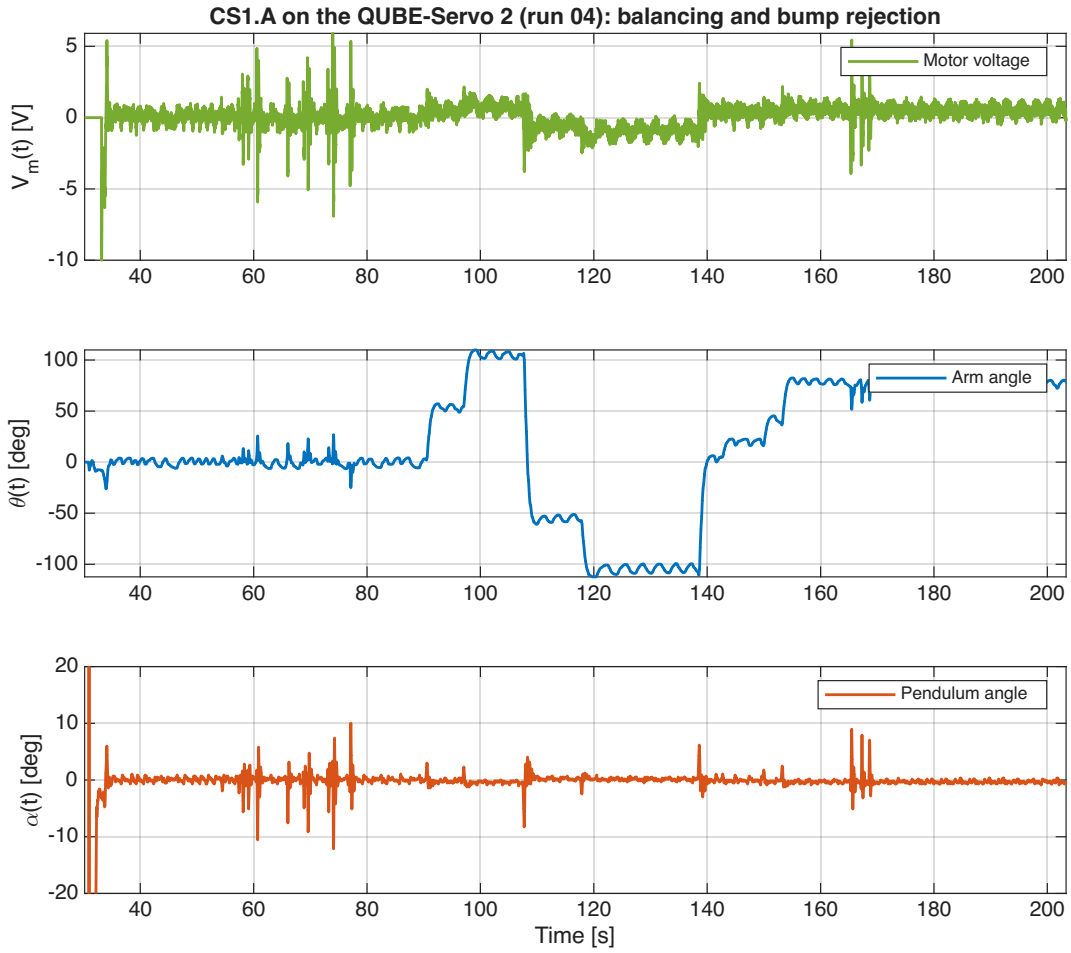


Figure 8: CS1 running on the QUBE-Servo 2 (run 04). The pendulum is caught at the start of the trace and then balanced for 171 s, during which it was bumped by hand several times.

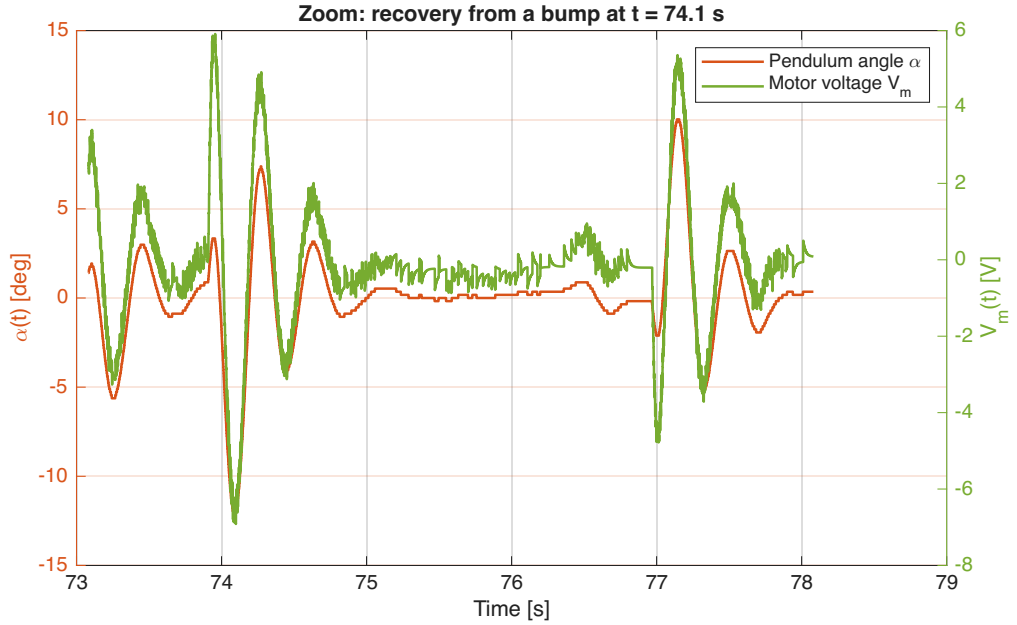


Figure 9: Zoom on Figure 8 around the largest bump: pendulum angle (left axis) and motor voltage (right axis).

Figure 8 shows CS1 balancing on the hardware for 171 s:

- **Balancing.** Excluding the catch transient, the pendulum angle has a standard deviation of $0.97°$, about five encoder counts, and an RMS voltage of 0.79 V.
- **Bump rejection.** The pendulum was bumped by hand several times. The largest bump tilted it by $12.1°$, and Figure 9 shows it back within $\pm 2°$ in about 3.5 s. The voltage briefly reaches about 5 V, and the response is well damped, as the $M_s = 1.66$ of Figure 4 predicts.
- **Voltage chatter.** The persistent ± 1 V ripple in V_m comes from the derivative terms acting on the quantised encoder signals.
- **Arm drift.** The arm drifts slowly through a range of about $\pm 110°$. Without integral action, Coulomb friction lets the arm rest anywhere inside a dead band, and each bump moves that resting point. **[TODO:** Confirm what the operator was doing with the arm and the reference during run 04.**]**

[TODO: Record a hardware $45°$ arm step with CS1 (with the reference logged) and add its plot and its measured M_p, t_s and steady-state error here. Runs 01–04 contain balancing and bumps only, with no logged step.**]**

7 Control System 2: discrete-time state feedback with an observer

7.1 Control structure and why it was chosen

State feedback uses all four states, so it can place every closed-loop pole directly. CS1 could only place five poles through four gains, with the fifth fixed by the pole-sum constraint. The design is done entirely in discrete time on the ZOH model (A, B, C) of Section 4, so no further approximation is introduced at implementation. Three additions turn the DP2 SFC into a controller that works on the hardware:

- a **Luenberger observer**, because only θ and α are measured;
- **integral action** on the arm error, to remove the offset caused by friction;
- a **first-order reference filter**, to keep a $45°$ step inside the voltage limit.

The control law is

$$r_f[k+1] = r_f[k] + \tfrac{T_s}{\tau}\big(r - r_f[k]\big), \qquad \tau = 0.2\,\text{s}, \tag{62}$$

$$x_i[k+1] = x_i[k] + T_s\big(\theta[k] - r\big), \tag{63}$$

$$u[k] = \mathrm{sat}_{\pm 10}\Big[-F\big(\hat{x}[k] - x^*\big) - F_i x_i[k]\Big], \qquad x^* = [r_f\ 0\ 0\ 0]^T, \tag{64}$$

$$\hat{x}[k+1] = A\hat{x}[k] + Bu[k] + L_o\big(y[k] - C\hat{x}[k]\big). \tag{65}$$

7.2 Controllability and observability

The controllability matrix $\mathcal{C} = [B\ AB\ A^2B\ A^3B]$ and the observability matrix $\mathcal{O} = [C;\ CA;\ CA^2;\ CA^3]$ both have rank 4 (MATLAB `rank(ctrb(A,B))` and `rank(obsv(A,C))`). All closed-loop poles can therefore be placed, and the full state can be reconstructed from the two encoders. The open-loop discrete poles include $z = e^{9.95T_s} = 1.00999$, outside the unit circle, which confirms the unstable plant.

7.3 Pole selection

The poles are chosen in the s-plane from the specifications and mapped to the z-plane with $z = e^{sT_s}$. This mapping is exact for the ZOH model, since $A = e^{A_c T_s}$:

- **SFC.** A dominant pair with $\zeta = 0.7$ and $\omega_n = 8$ rad/s ($-5.6 \pm 5.71j$), plus -19.2 and -25.6. This is the same bandwidth as CS1. The DP2 poles ($\omega_n = 2.5$) met the specification in simulation, but they left the loop stable only while the motor delivered at least $0.87\times$ its modelled gain. The CS1 hardware runs showed the real effective gain is lower, so the faster set, which tolerates $0.29\times$, was adopted.
- **Integrator.** The pole is at -1 s^{-1}. It was first placed at -8, but that roughly tripled the arm gain, so a small bump saturated the voltage and the pendulum fell. At -1, the gains stay close to plain SFC and friction offsets are removed in about one second.
- **Observer.** The poles are at $-90, -95, -100, -105$ s^{-1}, about $4\times$ faster than the fastest SFC pole, so the estimate converges well before the controller acts on it. They are not made faster, because a faster observer passes more quantisation noise into the velocity estimates.

7.4 Gain calculation

- **SFC and integral gains.** F and F_i are computed together on the augmented system $A_a = \begin{bmatrix} A & 0 \\ T_s[1\,0\,0\,0] & 1 \end{bmatrix}$, $B_a = \begin{bmatrix} B \\ 0 \end{bmatrix}$ with `place(Aa, Ba, exp([ps p_int]*ts))`, giving

$$F = [-19.41\ 10.19\ 5 - .59\ 48.16], \qquad F_i = -14.922.$$

- **Observer gain.** By duality, it is $L_o =$`place(A', C', exp(po_s*ts))'`.

Broken at the plant input, the state-feedback loop has a gain margin band of 0.29–$44.8\times$ and a phase margin of 67.7°. These are wider than CS1, because state feedback uses the measured velocities rather than a filtered derivative.

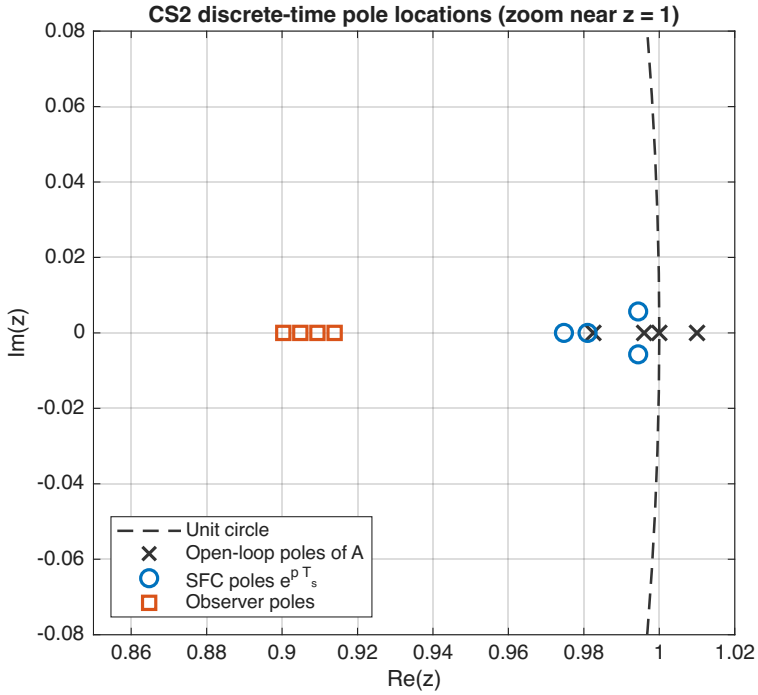


Figure 10: CS2 poles in the z-plane, zoomed near $z = 1$: open-loop poles of A (one outside the unit circle), the SFC poles and the observer poles.

Figure 10 shows the mapping $z = e^{sT_s}$ at work:

- the unstable open-loop pole sits just outside the unit circle;
- the SFC poles move it inside;
- the faster observer poles lie further from $z = 1$, because a larger $|s|$ maps to a smaller $|z|$.

7.5 Implementation

Like CS1, the whole controller is one MATLAB Function block (`ctrl_CS2.slx`) shared by both simulations and the hardware. It includes three practical details:

- **Bumpless start.** On enable, the observer starts from the measured angles with zero velocities, r_f starts from the measured θ, and $x_i = 0$, so the controller starts without a voltage jump.
- **Conditional integration with anti-windup.** The integrator updates only while $|\theta - r| \leq 2°$, and not while the voltage is saturated in the direction that would increase it. Integrating during the whole $45°$ move produced 44% overshoot in simulation.
- **Observer driven by the saturated voltage.** The observer uses the limited u, so it models the input the motor actually receives.

```
u_unlimited = -F*(xh - x_ref) - Fi*xi;
u = min(max(u_unlimited, -u_max), u_max);
e = theta - r; dxi = ts*e;
winding_up = (u_unlimited > u_max && -Fi*dxi > 0) || (u_unlimited < -u_max && -Fi*dxi < 0);
if abs(e) <= int_band && ~winding_up && ~reset, xi = xi + dxi; end
xh = A*xh + B*u + Lo*(y - C*xh); % Luenberger observer
```

7.6 Simulation results

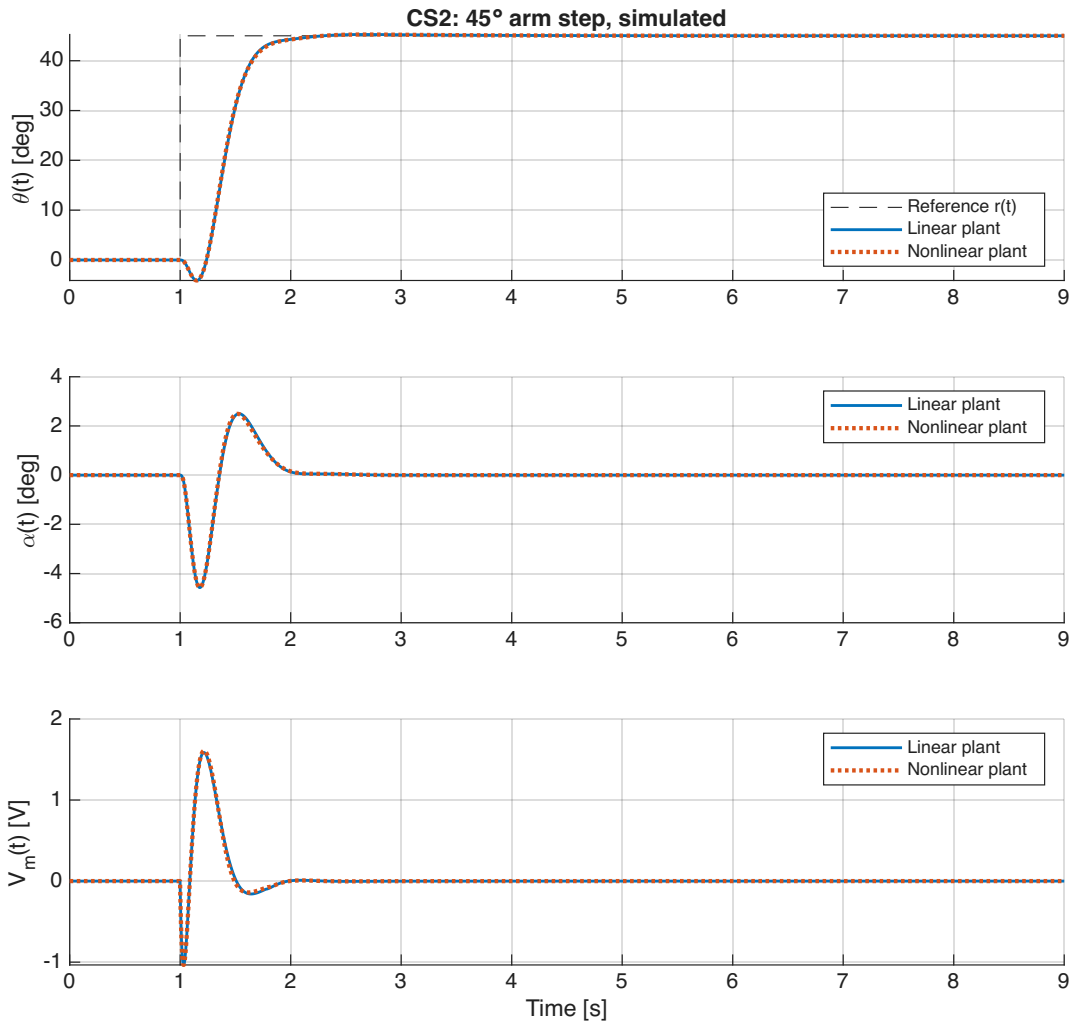


Figure 11: CS2 simulated response to a 45° arm step at $t = 1$ s on the linear and nonlinear plant models.

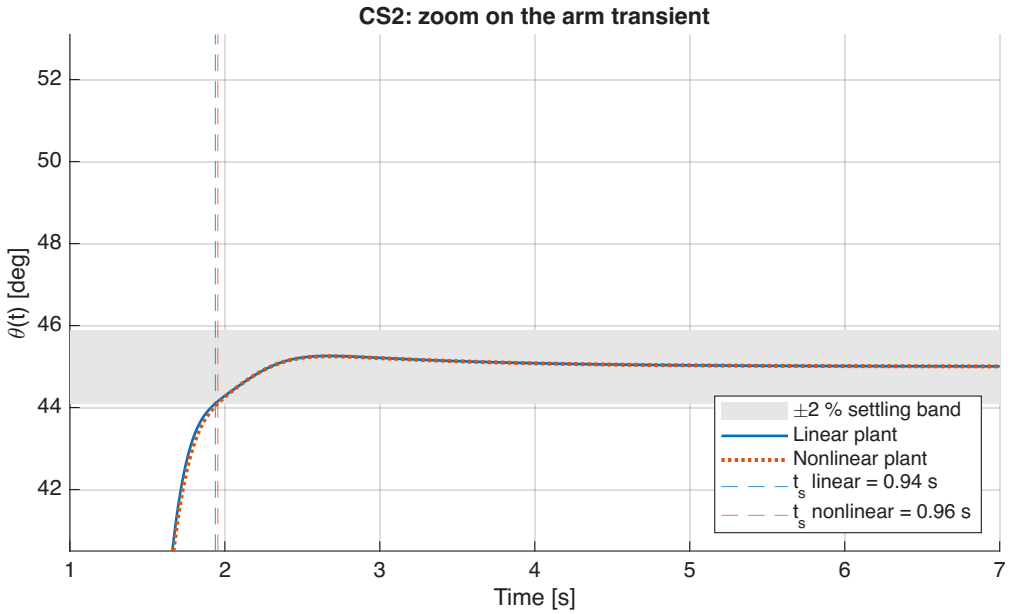


Figure 12: Zoom on the CS2 arm transient of Figure 11, with the $\pm 2\%$ settling band.

CS2 settles in $t_s = 0.96$ s with $M_p = 0.6\%$ on the nonlinear plant (Figures 11 and 12). This is faster than CS1, because the reference filter ($\tau = 0.2$ s) is lighter than CS1's prefilter. The cost is a higher peak voltage (1.6 V) and a larger pendulum excursion (4.6°), both still well inside their limits.

7.7 Experimental results

CS2 balances the pendulum on the QUBE. The first hardware version fell over when bumped, which led to the integrator-pole change described above. **[TODO:** Add the CS2 hardware run: balance, bump and 45° step with the reference logged, with its plot and measured M_p, t_s and steady-state error. No CS2 hardware data is saved in the repository yet.**]**

8 Comparison of the two control systems

Table 4: Performance for a 45° arm step. Simulation values are from `make_figures.m`; hardware values are still to be measured.

	CS1 (frequency domain)		**CS2 (SFC + observer)**	
	Nonlinear sim.	Hardware	Nonlinear sim.	Hardware
Overshoot M_p [%]	0.0	**[TODO:]**	0.6	**[TODO:]**
Settling time t_s [s]	1.64	**[TODO:]**	0.96	**[TODO:]**
Peak $\lvert V_m \rvert$ [V]	0.8	**[TODO:]**	1.6	**[TODO:]**
Peak $\lvert \alpha \rvert$ [deg]	2.4	**[TODO:]**	4.6	**[TODO:]**
Gain margin band	0.53–68	–	0.29–44.8	–
Phase margin [deg]	34.4	–	67.7	–

[TODO: Discuss Table 4 once the hardware columns are filled: which controller tracks better, which rejects bumps better, voltage chatter, and the friction offset (CS2 has integral action, CS1 does not).**]**

9 Response to feedback from the final demonstration

[**TODO:** List each comment or question from the teaching team at the final demonstration, what the team changed or tested in response, and the result with a figure or number. This section is marked as its own criterion.]

10 Conclusions and future work

Two control systems were designed from the same physically derived model of the QUBE-Servo 2 and implemented as single MATLAB Function blocks at 1 kHz.

CS1. Two filtered PD controllers with gains computed from specification-driven pole targets were verified by Bode, root-locus and sensitivity analysis. They have a 34.4° phase margin and tolerate a motor gain down to $0.53\times$ nominal. In simulation, CS1 meets the requirements with no overshoot and $t_s = 1.64$ s. On the hardware, it balanced the pendulum for minutes and recovered from hand bumps.

CS2. The discrete SFC with integral action and an observer meets the requirements in simulation, with $M_p = 0.6\%$ and $t_s = 0.96$ s. Its integral action removes the friction offset that CS1 leaves.

Key lessons.

- **Robustness as well as nominal specifications.** Designs that met the specifications on the nominal model failed on the hardware or under modelled quantisation. Bandwidth had to be chosen for robustness to the real motor gain and the encoder resolution.
- **One shared code path.** Keeping the simulated and implemented code identical made hardware debugging a matter of physics, not code.

Future work.

1. Identify the motor gain and Coulomb friction of this QUBE experimentally and add friction compensation.
2. Add integral action to CS1.
3. Design CS1 directly in the z-domain and compare it with the s-domain design.
4. Add an energy-based swing-up [3] so that the system starts from hanging down without manual help.

[**TODO:** Revise once hardware step results and demonstration feedback are added.]

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